# Win-Dis: Ontological Modelling of Window Connections to support Disassembly Planning

Fritz Trede[1,2*]

[1]Master Candidate, Digital Engineering M.Sc., Bauhaus University Weimar
[2]Visiting Student, Chair of Systems Engineering, Technical University Berlin

**Abstract**

*Windows are technically complex assemblies that vary in material composition, glazing types, and installation methods, and their disassembly is often constrained by hidden or irreversible connections. This paper presents WinDis, an ontology prototype that formalizes knowledge about window assemblies, their components, and the connections that hold them in place, as well as the processes and tools required for their removal. The ontology integrates physical, processual, and reuse-related perspectives, enabling queries about access, reversibility, residue risks, disassembly sequences, and reuse options. Verification with a reasoner confirmed logical consistency, and validation against competency questions served as a proof of concept and demonstrated that the model can answer key planning queries. Overall, WinDis provides a structured and extensible framework for representing disassembly knowledge and aims to provide a step toward more systematic reuse of window components in practice.*

## Motivation

Today, renovation projects constitute the majority share of construction projects, already accounting for 57% of the overall construction activity, despite an annual renovation rate of only 1–2% [1].

This share is expected to increase due to driving factors such as rising energy costs and EU sustainability targets, which together render the operation of inefficient buildings increasingly uneconomical. At the same time, the steadily increasing performance requirements of renovation products in recent decades have made them progressively more complex [1, 2].

In this context, windows provide a clear example. Their assemblies have become progressively more intricate — both in terms of materials and composition (e.g., insulated frames, gas-filled cavities) [2, 3]. On the other hand, recent legislation at the EU and member-state level requires planners not only to adhere to performance standards but to also consider circularity principles such as ease of disassembly, reuse, and recycling [4].

In practice, window disassembly remains challenging, especially at scale. Planners must answer practical questions: How are components connected? Can they be accessed and removed without damage? What sequence and tools are required? Furthermore, since connections are often obscured by layers of substrate, the removal process is uncertain and frequently destructive. Alongside the disassembly process itself, these challenges negatively impact the planning process itself, making it time- and cost-intensive. Although initiatives exist to recycle materials such as glass, aluminium, and PVC, uptake remains limited and is usually pursued only when economically viable [5].

In this context, shared knowledge representations may offer a way forward. They can help provide useful knowledge to support decision-making in different deconstruction scenarios (e.g., selective glazing replacement, unit removal, or full façade renovation). By providing structured information, such representations can reduce the need for destructive inspections and mitigate uncertainty.

Building on this motivation, the remainder of this paper is structured as follows: Section  reviews relevant norms, guidelines, and ontological resources. Section  outlines the research design, the ontology development approach and specification Section **??** introduces the conceptual design of the ontology. Section  describes implementation, verification, and validation process. Section **??** discusses the outcomes, limitations, and future work and Section  summarizes key takeaways.

## Background

Understanding disassembly requires first examining window assemblies in their physical and functional composition. Within the EU, DIN EN 14351 provides a harmonized standard for windows and external doorsets. It defines performance characteristics such as air permeability, watertightness, wind resistance, acoustic and thermal insulation, and durability, alongside associated test methods. Although it does not fully specify complete assemblies, the standard distinguishes functional components such as framing

*Corresponding author: FTrede@gmail.com

elements, glazing, hardware, and weatherstripping. It also indicates which modifications in a window assembly require renewed performance testing [6].

The RAL Guideline for Window and Door Installation (RAL-Montageleitfaden) complements DIN EN 14351 by providing detailed guidance on assembly and installation practices of punched windows. According to the guideline, planning the dismounting and disassembly of existing windows constitutes a dedicated step in the renovation process. Although developed in Germany, the guideline is available in English and is referenced internationally. It describes common connection details and installation practices, including sealing systems, beads, and foams, and explains how these interact with typical wall opening characteristics encountered in practice. While its main focus lies in new construction, it also addresses renovation and disassembly, highlighting how different installation methods can facilitate or complicate later removal. In this context, the guideline distinguishes five window renewal scenarios:

- Modernization: The window exchange occurs in conjunction with a complete façade overhaul.
- Exchange of complete unit: Removal and replacement of the entire window including reveal.
- Exchange of window including frame: Removal of the window and frame, followed by installation of a new unit in the wall opening.
- Using old frame as reveal: Retention of the old frame, into which the new frame is inserted.
- Two-step window subframe: A new frame is mounted into an existing subframe of the old window.

Furthermore, the guideline lists typical hazardous materials in window assemblies, their locations, and the construction periods in which they are most likely to occur [7].

However, for deconstruction planning, understanding assemblies alone is insufficient. The separation process itself must also be considered, including accessibility, sequencing of operations, and required tools.

The Deconstruction Ontology (DCO), based on DIN EN 18007, offers a framework for knowledge-based deconstruction. It categorizes methods such as sawing, blasting, and ablating, alongside required equipment and component characteristics (e.g., load-bearing capacity, hazardous content). Yet, since it targets construction products in general and does not model deconstruction processes, its applicability to the specific case of window disassembly is limited [8].

Another relevant source is Reno-Inst, which particularly addresses renovation planning. It models typical renovation products, including windows, ETICS, and HVAC systems, and supports renovation activities by representing installation processes, sequencing, and constraints. While Reno-Inst provides a comprehensive resource for renovation projects, its emphasis lies primarily on construction rather than deconstruction and disassembly. Furthermore, although it lists window components, it does not explicitly model connection details, which are critical when assessing how a complex product can be disassembled into constituent parts [9].

In general, ontological modelling of domain knowledge has become a well-established approach in the Architecture, Engineering, and Construction (AEC) industry. To date, numerous ontologies have been introduced that capture building assets at different levels of granularity and represent associated stakeholder processes. Examples include the Building Topology Ontology (BOT), which models building elements, spaces, and relationships; the Building Product Ontology (BPO), which describes product properties; and the Material Passport Ontology, which captures physical, functional, and environmental characteristics [10, 11]. However, the majority of these ontologies primarily model buildings and construction processes aimed at new construction. By contrast, the DCO and Reno-Inst represent novel efforts to model deconstruction and renovation processes [9, 8].

## Gap and Contribution

With regard to the specific requirements of window disassembly, existing ontological resources lack the necessary granularity and specificity to capture the processes and considerations relevant for planning. Windows are intricate assemblies that are both technically complex and frequently replaced in renovation projects. Their disassembly therefore requires particular attention. Better planning of disassembly activities can increase efficiency, raise awareness of end-of-life options, and enhance the potential for recycling and reuse.

This study addresses this gap by developing a dedicated ontology of window assemblies (i.e. punched windows) and their immediate surroundings, capturing physical, functional, and compositional characteristics with a particular emphasis on component connections. The ontology is designed to support the modelling of disassembly processes, thereby assisting deconstruction planners in scoping and planning practices. To date, no unified framework exists for the disassembly of window products; only general principles and methods are available. This research thus aims to provide a novel, domain-specific resource for window disassembly planning.

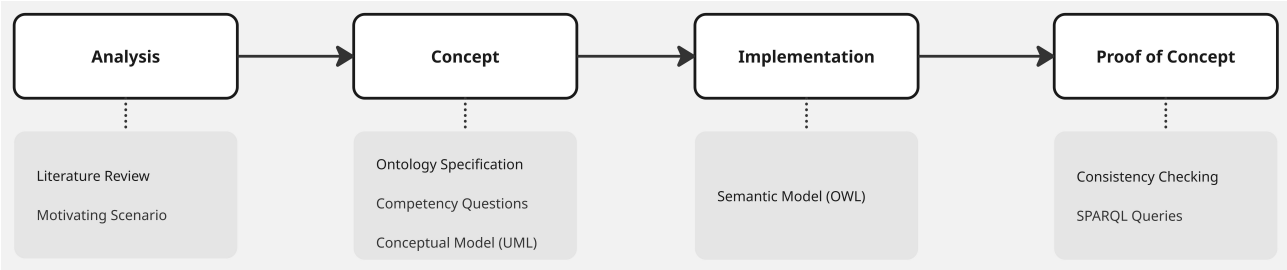


**Figure 1:** *Research Approach*

# Research Design

This section introduces the research approach applied in developing the WinDis Ontology. As shown in Figure 1, the study follows an analysis – concept –implementation – proof of concept approach. While previous sections presented the literature review and motivation, the following section outlines the activities and principles guiding the ontological modeling process before detailing the ontology specification and competency questions.

The conceptual modeling followed the guidelines of Noy and McGuinness [12]. The process began by defining the ontology's domain and scope, then identifying relevant concepts and their properties. From this specification, a set of competency questions was derived to guide development and prevent scope creep.

Initially, the ontology was intended to be developed using a model-driven architecture (MDA) paradigm. Due to limited tool support, this approach was abandoned. As a result, visualizations presented in the next section use Unified Modelling Language (UML) syntax, while in subsequent sections the formal representation uses Web Ontology Language (OWL) terminology.

For the implementation, the identified concepts were formalized as OWL constructs using the open-access tool Protégé. To this end, classes, individuals, object-, and data-properties were created to represent the knowledge base.

Finally, the proof-of-concept stage essentially addressed two questions: (1) Was the ontology built correctly? (Verification) and (2) Was the right ontology built? (Validation). Verification was carried out with Protégé's built-in reasoning tool (HermiT). Validation was performed by populating the ontology with example instances and testing functionality through SPARQL queries derived from the competency questions.

## Ontology specification

The aim of the ontology specification is to develop a general description of the ontology regarding its domain, scope and users. Here, the specification will be presented using natural language and a set of competency questions.

1. *What is the purpose of the ontology?*

   The ontology aims to provide a structured, machine-interpretable representation of windows and their assemblies, specifically focusing on their compositional characteristics, connections between them and the processes required for their disassembly. Its purpose is to support deconstruction planning by making implicit and scattered knowledge explicit, accessible, and reusable.

2. *What is the scope of the ontology?*

   The ontology models window products and their immediate surroundings (i.e. wall openings), including: Physical components of windows, types of connections and fastening methods, material composition, disassembly activities, their sequences, tools, and constraints. It focuses on renovation and deconstruction use cases rather than new construction or generic building components.

3. *Who are the intended end-users?*

   Deconstruction planners, architects and engineers involved in renovation projects, and researchers and ontology developers in the AEC domain.

4. *What is the intended use?*

   The ontology is intended to serve as a knowledge base and modeling support for disassembly planning of windows. It aims to support decision-making by providing structured information about components, connections, and tools. Figure 2 illustrates the potential use for disassembly planners. Potential benefits of the ontology include: Reduction

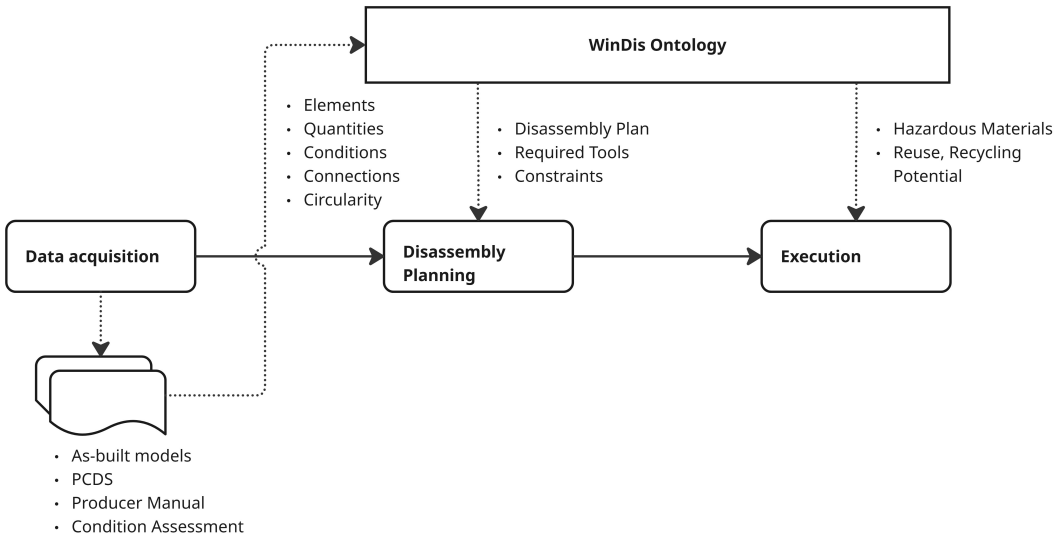


**Figure 2:** *WinDis Ontology - Use*

of destructive condition assessments and sampling; Facilitation of reuse and recycling by enabling planners to design selective, efficient disassembly processes. To sharpen focus in the development process and to avoid scope creep, a set of competency questions was defined, these can be seen in table **??** alongside example answers.

# Conceptual Model

This section introduces the conceptual model of the ontology, providing an overview of the key concepts and their relations. The concepts were derived from the literature review, with Table 2 indicating the specific sources that informed their identification. Figure 3 illustrates the ontology's key concepts and their relationships using UML notation. The following subsections introduce the key concept in more detail.

## Overview

The WinDis ontology is conceptually structured into five branches. First, the Physical Element, it describes what the window physically is. It describes the window assembly, its components and the wall opening it is installed in. Second, an Attribute provides information on a the physical and functional characterists. Third, a Connection describes how parts are joined: the way they are fastened or bonded, if they are reversible as well as the interfaces that decide connection accessibility. Fourth, the process model describes how window components can be dismounted from the wall opening and how they can be disassembled, according to a set of scenarios. It models disassembly plans, steps, their sequence, as well as required tools and preconditions. A fifth category "Other" was introduced to group further key considerations which inform disassembly planning such as reuse potential and barriers, and condition assessments.

## WinDis:PhysicalElement

In the `PhysicalElement` Class, a `WindowAssembly` represents a concrete window unit *in situ*. Its use is to group a set of components that together are installed as a window unit inside a `WallOpening`. It shall be noted here that connections are modeled on a component level, hence a `WindowAssembly` gathers the set of connections of its components. To this end, it draws components from a set of component families. First, a `FrameSystem` represents the load-bearing and operable perimeter: `Frame`, `Subframe` (if present), and `Sash` (for operable windows). They carry hardware and transfer loads to the opening. Next, the `GlazingSystem` is the transparent unit sitting within the `Frame` or `Sash`. A `RetentionAndSealSystem` describes elements that hold and seal glazing and regulate air and water, such as beads, gaskets and weathersealing. On the other hand, `Hardware` such as hinges or brackets join multiple components, or contribute to the window operation (e.g. locks). Overall, by representing the parts that must be handled, separated, or preserved, this class provides the material foundation for all planning decisions.

## WinDis:Attribute

`Attributes` describe measurable or descriptive properties of a `PhysicalElement` (i.e. window assemblies, components or wall openings). They hold information that do not define a components identity. Exam-

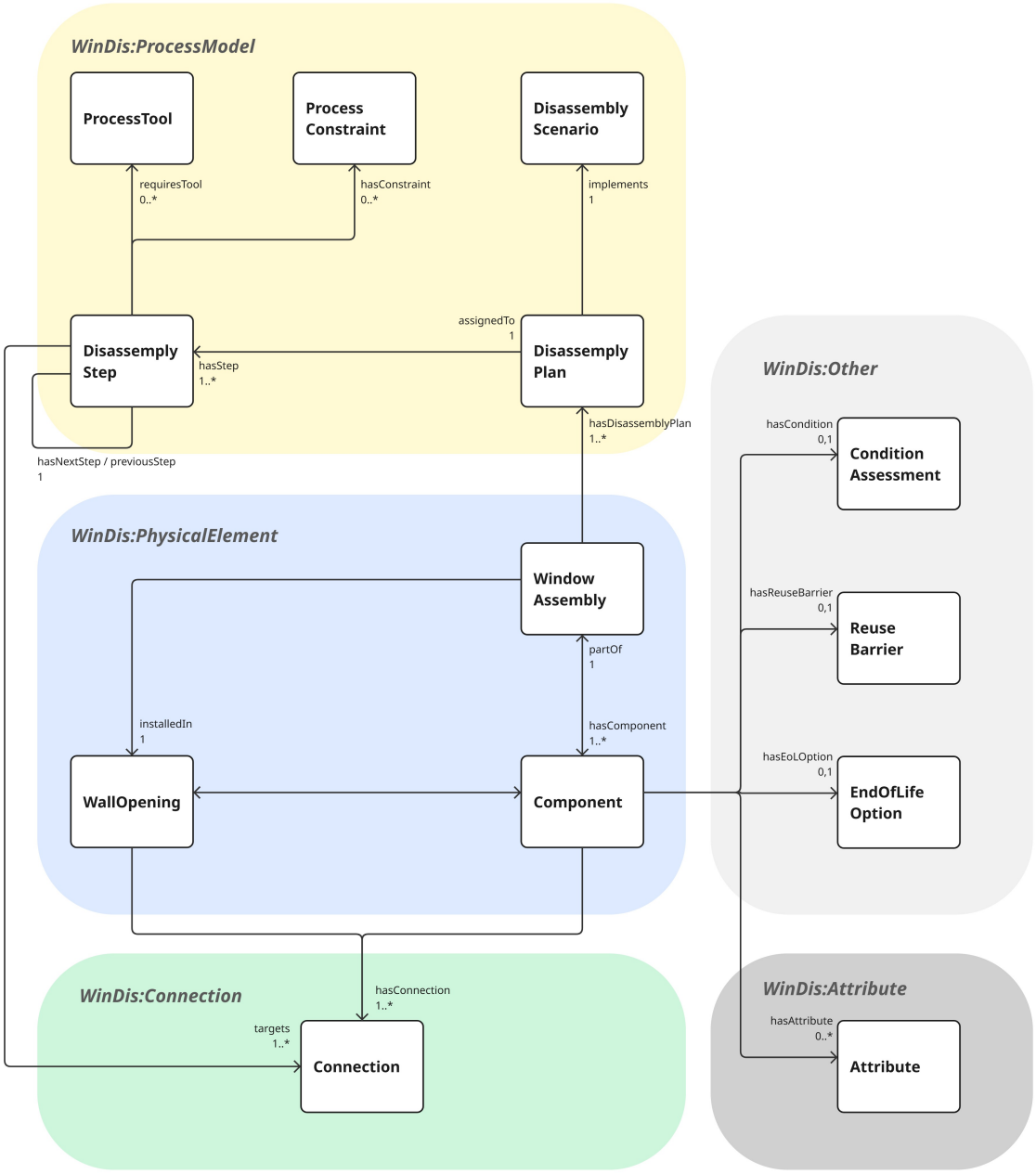


**Figure 3:** *WinDis ontology - Overview (UML)*

**Table 1:** *Competency questions with example answers*

| Competency Question | Example Answer |
|---|---|
| What materials make up the window's components? | Subframe, Frame, Sash (Timber); Double-Pane Float Glass; Sealants (Silicone) |
| Which side provides access to glazing beads or fasteners? | Glazing beads (Interior); Fasteners (Exterior, hidden behind sealing system) |
| How is the window frame connected to the surrounding wall? | Anchor brackets; Expanding foam insulation. |
| Which connections are reversible and which are destructive? | Screwed brackets (reversible); polyurethane foam (destructive) |
| What tools are required to release each connection type? | Screwdriver (Brackets); Oscillating knife (Foam); Pry bar (Beading) |
| What is the correct sequence of disassembly activities for this window type? | 1. Remove glazing beads; 2. Extract glazing unit; 3. Unscrew brackets; 4. Cut insulation foam; 5. Detach frame. |
| What are the preconditions for each activity? | Extract glazing unit (Precondition: Remove Glazing Beads) |
| Are hazardous materials present in the window assembly? | Lead paint; Asbestos |
| What is the current condition of each window component? | Glazing intact with condensation between panes; Gaskets brittle. |
| What is the residual service life of each component? | Frame: 10 years; Glazing: 15 years; Gaskets: expired. |

**Table 2:** *Main concepts, examples, and sources for ontology development*

| Main Concept | Term Examples | Sources |
|---|---|---|
| Physical Element | Sash, Glazing, Hardware, Wall Opening | [6, 7] |
| Attribute | Width, Height, UValue, Residual Service Life | [7, 6] |
| Connection | AdhesiveBond, MechanicalFastener, Access Side | [7, 8] |
| Process Model | Disassembly Step, Process Tool | [7, 9] |
| Other | Residual Service Life, Condition Assessment | [7, 4, 3, 10] |

ples include dimensional values (e.g. width, height, thickness), performance indicators (e.g. U-Value), or time related aspects such as residual service-life. In practice, these characteristics can help determine how heavy a unit is to lift, whether it can be reused, and how much effort is required to separate it, making them essential for safe and effective planning.

## WinDis:Connection

Modeling connections is at the core of the WinDis ontology, since understanding how components are connected to each other ultimately determines how they can be separated.

Each `Connection` relates to exactly two parts, a host or base part and an attached or fastened part. For every connection, the ontology records several aspects. It specifies the mechanism type, specifying how two parts are joined together, for example bolted. Furthermore, it records the access side, indicating whether the connection can be reached without destructive works. It characterizes reversibility, meaning whether the joint can be undone without damage. It also describes the residue risk, which refers to the likelihood of contaminating residues remaining after separation. In addition, the ontology captures the connection location, identifying where the joint is situated along the frame or opening interface, and the coverage, distinguishing between a connection at a single point and one that extends across a larger area such as the entire perimeter.

## WinDis:ProcessModel

Knowing a windows parts and their connection is not enough — as planning requires a sequence of actions. To this end, within the `ProcessModel` a `DisassemblyPlan` aggregates a se-

quence of DisassemblySteps, each targeting a specific Connection, with the ability to declare preconditions. Furthermore, a DisassemblyPlan can be assigned to a DisassemblyScenario (e.g. modernization), and can also reference specific ProcessTools (e.g. suction cups, silicone cutter) or constraints (e.g. safety, environmental, occupant, access). This provides the operational logic, allowing planners to turn static knowledge into an actionable procedure.

### WinDis:Other

The Other branch of the ontology captures decision-support factors that sit adjacent to the step-by-step plan. These factors do not directly affect the disassembly procedure itself, but they strongly influence decision-making and thus motivate disassembly activities.

One element is the ConditionAssessment, which records findings relevant to disassembly and reuse. Examples can include notes such as "frame heartwood intact; local rot at sill scarf" or "seal failure on IGU 2,". Furthermore, a ConditionAssessment may also include the residual service life of a component.

Another element is the ReuseBarrier, which highlights factors impeding reuse such as hazardous finishes, or non-standard sizes, and which may instead steer decisions toward recycling or selective salvage.

Finally, the ontology records EndOfLifeOptions, capturing the intended pathways for components — direct reuse, remanufacture, material recycling, or energy recovery — based on both the component condition and its separability.

### Example Scenario

An example reasoning loop supported by the ontology begins with the identification of the object of work, typically a WindowAssembly situated in a WallOpening.

From there, the planner analyses the joinery, mapping all relevant Connections of the assembly and the wall, while noting their MechanismType, AccessSide, Reversibility, Location, and Coverage.

The next step is to shape a DisassemblyPlan by sequencing DisassemblySteps so that reversible and accessible connections are released first — for instance, removing beads, extracting glazing, and detaching the sash — before tackling covered or destructive ones such as perimeter foam cutting or frame unseating. The feasibility of the plan can then checked by mapping Tools and Constraints to each step. An could include restrictions on the use of heat guns near historic varnish or the requirement of a two-person lift for heavy components.

Finally, the planner reviews the ConditionAssessment, ReuseBarriers, and EndOfLifeOptions to steer the plan toward reuse where warranted and to anticipate potential hazards.

## Implementation and Proof of Concept

This section introduces the translation of the conceptual model into an OWL ontology. The ontology was authored in OWL 2 (Turtle) using the open-access tool Protégé and its built-in features. The following paragraphs outline the main design decisions taken during implementation. Not all classes and properties are discussed in detail; the full ontology is accessible at https://github.com/wrldedit/windis-ontology, and readers are referred there for reference.

The ontology was developed under the *open world assumption* (OWA), which means that what is not stated cannot be assumed false but only unknown. This assumption seemed sensible, as windows, and their assemblies, exhibit many variations, and it is unrealistic to anticipate every possible case, especially in the domain of window disassembly. However, in specific instances a partial closed world assumption was applied. For example, when modeling the accessibility of a connection, it is reasonable to assume that this can only be *Inside*, *Outside*, *Both*, or *None*. In such cases, exhaustive enumeration was used to enforce closure.

### Class Structure

The backbone of the ontology is provided by the conceptual model. Key concepts and their children were implemented as classes and subclasses, following the structure shown in Figure 4. This ensures alignment between the conceptual reasoning about window disassembly and its semantic implementation.

### Object Properties

Relations between classes were implemented as object properties. Within **PhysicalElement**, hasComponent/partOf express assembly decomposition (with partOf declared transitive), while installedIn links a fenestration assembly to its wall opening. For **Connections**, hasConnection relates assemblies or components to the joins that hold them. Each **Connection** specifies exactly one fastenedPart (the attached component) and at most one basePartComponent and/or basePartWall (the host), thereby supporting both internal joins (e.g.,

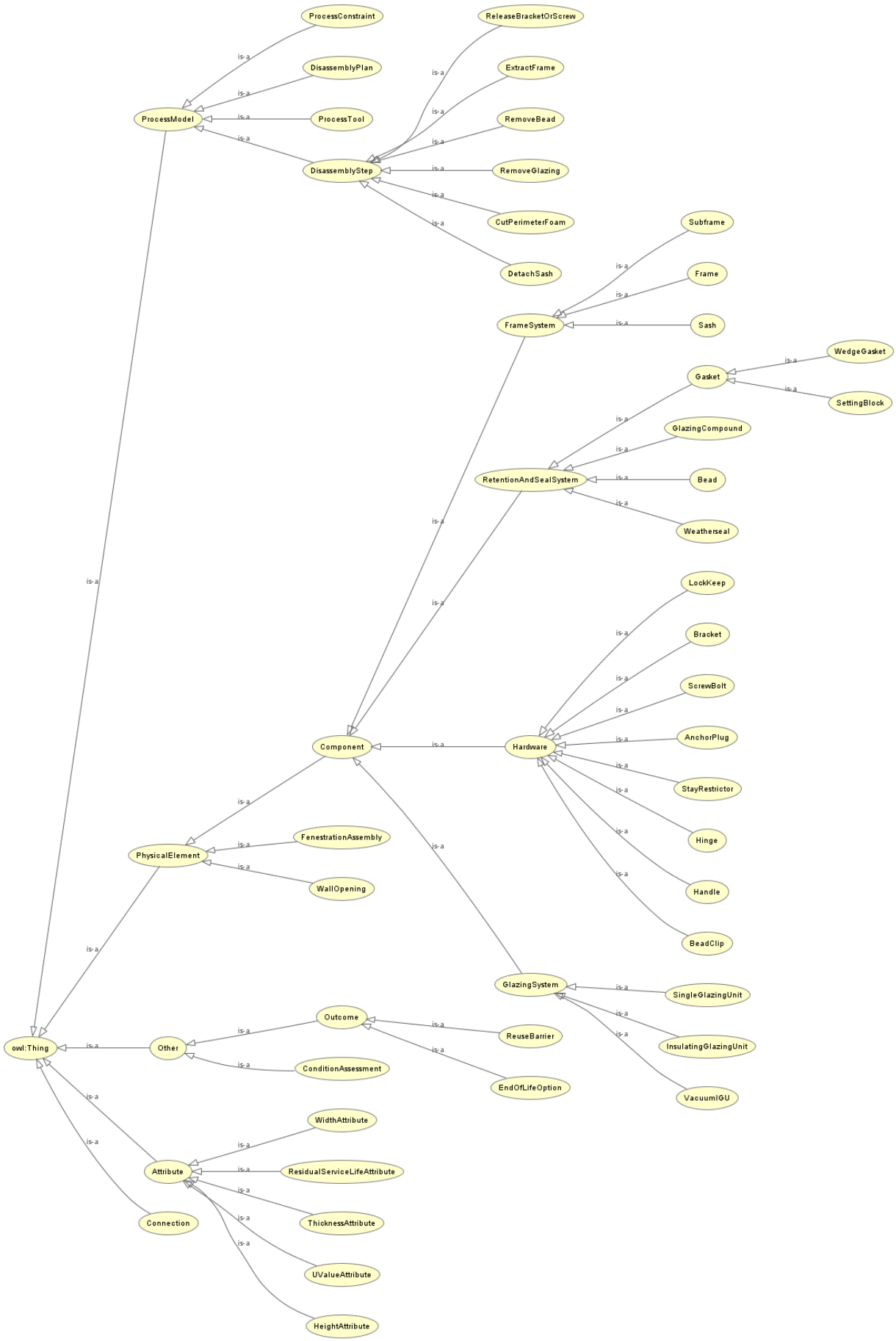


**Figure 4:** *WinDis Ontology - Class Hierarchy*

glazing to sash) and installation joins (frame to opening). The process domain uses `hasDisassemblyPlan`, `hasStep`, `hasPrecondition`, `hasNextStep`, `targetsConnection/targetsComponent`, and `usesTool` to represent the ordering of activities and anchor each step in what is being released and how.

### Data Properties

The semantic model differentiates data properties according to the values they take and whether they are composite in nature:

1. **Boolean flags.** Simple features, such as whether an assembly contains hazardous materials or whether a wall opening has a cover overlap, were modeled as `xsd:boolean`. Naming follows typical Boolean prefixes (e.g., `hasCoverOverlap`, `containsHazardousMaterial`).
2. **Descriptive fields.** In some cases stakeholders may need to record open-text information without analytical constraints. For example, `assessmentNote` allows short descriptive condition statements.
3. **Categorical choices.** Where closed categorical values are appropriate, datatype properties constrained to explicit code lists were used. Examples include `accessSide` ({Inside, Outside, Both, None}), `connectionLocation` ({Jamb, Head, Sill, Mullion}), and `coverage` ({Local, Perimeter}).
4. **Measured or declarative values.** Some values require explicit units and provenance. These were modeled using an **Attribute** pattern. Attributes are individuals attached via `hasAttribute`, each carrying `hasValue/hasUnit` and optionally `observedAt/observedBy`. This supports multiple observations over time, heterogeneous units, and source tracking.

### Verification

Logical verification was carried out using Protégé's built-in HermiT reasoner. Consistency checks ensured that all classes were satisfiable and that relations such as `partOf` behaved transitively without producing cycles.

### Validation

Validation was performed by translating the competency questions into SPARQL queries. These queries tested key aspects of the model: materials and finishes of components; accessibility and reversibility of connections; frame-to-wall connection types; required tools; disassembly sequences and preconditions; hazard presence; and residual service life. To enable meaningful testing, the ontology was populated with two contrasting examples: a timber assembly with an argon-filled IGU and interior beads, and an aluminum assembly with a vacuum IGU and exterior beads. Both examples included frame-to-wall fixings (reversible screws/brackets and destructive foam), internal glazing retention, bead accessibility (inside vs. outside), and complete disassembly plans with ordered steps and preconditions. Condition assessments with residual service life were attached to key components to demonstrate decision-support queries. Each query produced the expected results by returning specific class types for readability. Together, these runs confirmed that WinDis can answer the intended questions. Specifically, what is held to what, where it is accessible, how it is released, in which order, with which equipment, and to what reuse-oriented outcome it might be steered. The queries and example instances are available in the GitHub repository. Figure 5-6 illustrate excerpts from the SPARQL queries.

## Discussion

This section reflects on the outcomes of the study. It revisits the knowledge acquisition process, discusses the semantic modelling approach and its limitations, and concludes with directions for future work.

### Knowledge acquisition and validation

Knowledge acquisition proved challenging. Available sources mainly addressed the physical composition and functional requirements of window assemblies alongside their installation, while design-for-disassembly practices are not yet widely established in industry. Although research exists in areas such as glass recovery and recycling, systematic knowledge on the practicalities of window disassembly is scarce. For this work, installation manuals and guidelines were consulted, and disassembly scenarios were inferred from installation practices. Consequently, the conceptualisation relied heavily on literature review and logical reasoning rather than empirical observation. It must also be acknowledged that the author is not a domain expert in disassembly planning or façade engineering. The ontology was developed within the scope of a 180-hour project and has not been validated with practitioners, which limits its reliability and practical relevance. Expert validation therefore represents a crucial next step.

### Semantic model

The ontology implementation followed a clear structure, distinguishing physical elements, connections,

SPARQL query:

```
PREFIX windis: <http://example.org/windis#>

SELECT ?connection ?location ?coverage ?rev ?residue
    (GROUP_CONCAT(DISTINCT ?toolName; separator=", ") AS ?tools)
WHERE {
 ?connection a windis:Connection ;
        windis:fastenedPart windis:Frame1 ;
        windis:basePartWall windis:WO1 .
 OPTIONAL { ?connection windis:connectionLocation ?location }
 OPTIONAL { ?connection windis:coverage ?coverage }
 OPTIONAL { ?connection windis:reversibility ?rev }
 OPTIONAL { ?connection windis:residueRisk ?residue }
 OPTIONAL {
  ?connection windis:requiresTool ?t .
  BIND(COALESCE( STRAFTER(STR(?t), "#"), STRAFTER(STR(?t), "/")) AS ?toolName)
 }
}
GROUP BY ?connection ?location ?coverage ?rev ?residue
ORDER BY ?connection
```

| connection | location | coverage | rev | residue | tools |
|---|---|---|---|---|---|
| C_FrameFoamToWall | | "Perimeter"^^<http://www.w3.org/2001/XMLSchema#strin | "Destructive"^^<http://www.w3.org/2001/XMLSchema#strir | "High"^^<http://www.w3.org/2001/XMLSchema#string> | "" |
| C_FrameToWall_Head | "Head"^^<http://www.w3.org/2001/XMLSchema#string> | "Local"^^<http://www.w3.org/2001/XMLSchema#string> | "Reversible"^^<http://www.w3.org/2001/XMLSchema#strir | "None"^^<http://www.w3.org/2001/XMLSchema#string> | "" |
| C_FrameToWall_Jamb | "Jamb"^^<http://www.w3.org/2001/XMLSchema#string> | "Local"^^<http://www.w3.org/2001/XMLSchema#string> | "Reversible"^^<http://www.w3.org/2001/XMLSchema#strir | "None"^^<http://www.w3.org/2001/XMLSchema#string> | "" |

**Figure 5:** *WinDis Ontology - Example SPARQL Query - Connections*

SPARQL query:

```
PREFIX windis: <http://example.org/windis#>
PREFIX rdfs:   <http://www.w3.org/2000/01/rdf-schema#>

SELECT ?component ?typeName ?material ?finish ?hazardFlag
    (GROUP_CONCAT(DISTINCT ?substance; separator=", ") AS ?substances)
WHERE {
 windis:FA1 windis:hasComponent ?component .

 # most-specific (leaf) type
 ?component a ?type .
 FILTER NOT EXISTS {
  ?sub rdfs:subClassOf ?type .
  ?component a ?sub .
  FILTER (?sub != ?type)
 }
 BIND(COALESCE( STRAFTER(STR(?type), "#"), STRAFTER(STR(?type), "/")) AS ?typeName)

 OPTIONAL { ?component windis:materialKind ?material }
 OPTIONAL { ?component windis:finishType  ?finish }
 OPTIONAL { ?component windis:hazardousOverall ?hazardFlag }
 OPTIONAL { ?component windis:substancesPresent ?substance }
}
GROUP BY ?component ?typeName ?material ?finish ?hazardFlag
ORDER BY ?component
```

| component | typeName | material | finish | hazardFlag | substances |
|---|---|---|---|---|---|
| Bead1 | | "PVC"^^<http://www.w3.org/2001/XMLSchema#string> | "FoilLaminate"^^<http://www.w3.org/2001/XMLSchema#st | | "" |
| Frame1 | | "Timber"^^<http://www.w3.org/2001/XMLSchema#string> | "Paint"^^<http://www.w3.org/2001/XMLSchema#string> | "true"^^<http://www.w3.org/2001/XMLSchema#boolean> | "Lead" |
| Gasket1 | | "PVC"^^<http://www.w3.org/2001/XMLSchema#string> | | | "" |
| IGU1 | | | | | "" |
| Sash1 | | "Timber"^^<http://www.w3.org/2001/XMLSchema#string> | "Paint"^^<http://www.w3.org/2001/XMLSchema#string> | | "" |

**Figure 6:** *WinDis Ontology - Example SPARQL Query Materials ans Hazards*

processes, attributes, and reuse-related considerations. Categorical features were expressed as enumerated datatypes, descriptive information as literals, and measurable values with units and provenance through an attribute pattern. This approach kept the model coherent and queryable. On the other hand integration with other ontologies was presents a crucial next step. Condition assessment could be aligned with defect taxonomies, while links to BIM schemas or national object type libraries could enhance interoperability. Theses choices result in a lean prototype which is open for future development.

## Future work

Further development could proceed along several directions. Validation with domain experts is essential to confirm that the conceptual model is relevant for the industry. Additional scenarios, such as modernization, glazing exchange, and gasket renewal, should be integrated more explicitly into disassembly plans. From a semantic perspective, alignment with existing ontological resources, for example for condition assessment or BIM-based object models would strengthen interoperability. Moreover, as European circular economy requirements evolve alongside research into recycling and reuse of building products - concepts for reuse barriers, end-of-life options, and material traceability will likely require extension. The modular architecture of WinDis provides a solid foundation for this incremental evolution.

# Key Takeaways

This project demonstrates that the knowledge surrounding the disassembly of window assemblies can be formalized in a coherent and queryable ontology. The resulting WinDis ontology captures both the physical composition of windows and the processual dimension of disassembly, with a strong emphasis on how components are connected and how these connections can be released. By modeling connection mechanisms, access, reversibility, residue risk, and location, the ontology is able to represent important factors that determine feasible and efficient disassembly sequences.

The work utilized a modular design strategy, distinguishing between physical elements, connections, process models, attributes, and reuse-oriented considerations. This has resulted in a structure that is flexible, and extensible. Verification and validation confirmed consistency and functionality of the ontology by utilizing the HermiT reasoner and translating the competency questions into SPARQL queries.

At the same time, limitations remain. The knowledge base was assembled from guidelines, manuals, and logical inference rather than direct domain exper-

tise, and condition assessment and reuse pathways are only treated in a simplified form. The ontology does not yet integrate with broader building information models or condition assessment ontologies, and such alignment will be needed for improved funtionality

Overall, the key contribution of this work is the demonstration that ontology-based modeling can support disassembly planning for windows in a structured, explicit, and reusable way. WinDis provides a foundation that can be extended and validated in collaboration with practitioners, and that can be developed to integrate with emerging circular economy frameworks and building information standards.

## References


[1] Ingeborg Artola et al. *Boosting Building Renovation: What Potential and Value for Europe?* Tech. rep. Study commissioned by the European Parliament's Committee on Industry, Research and Energy (ITRE). Policy Department A: Economic and Scientific Policy, European Parliament, June 2016. URL: https://www.europarl.europa.eu/RegData/etudes/STUD/2016/587326/IPOL_STU(2016)587326_EN.pdf.

[2] Bjørn Petter Jelle et al. "Fenestration of today and tomorrow: A state-of-the-art review and future research opportunities". In: *Solar Energy Materials and Solar Cells* 96 (2012), pp. 1–28. DOI: 10.1016/j.solmat.2011.08.010.

[3] Glass for Europe. *Glazing Potential in Germany*. https://glassforeurope.com/wp-content/uploads/2020/01/Glazing_potential_brochure_2019_German.pdf. Brochure on the energy savings and $CO_2$ reduction potential of high-performance glazing in Germany. 2019.

[4] Nicholas Dodd, Shane Donatello, and Mauro Cordella. *Level(s) indicator 2.4: Design for deconstruction user manual: introductory briefing, instructions and guidance (Publication version 1.1)*. Tech. rep. Available at https://susproc.jrc.ec.europa.eu/product-bureau/product-groups/412/home. Seville, Spain: European Commission, Joint Research Centre (JRC), 2021.

[5] Mathieu Hestin, Sarah de Veron, and Stephanie Burgos. *Economic Study on Recycling of Building Glass in Europe*. Tech. rep. Quantification of building-glass waste and analysis of recycling scenarios in the EU-28. Deloitte Sustainability, commissioned by Glass for Europe, Apr. 2016. URL: https://www.glassforeurope.com/wp-content/uploads/2018/04/Economic-study-on-recycling-of-building-glass-in-Europe-Deloitte.pdf.

[6] *DIN EN 14351-1:2016-12. Windows and doors - Product standard, performance characteristics - Part 1: Windows and external pedestrian doorsets without resistance to fire and/or smoke leakage characteristics*. German national adoption of EN 14351-1:2006+A2:2016. Berlin, Germany, 2016. URL: https://www.beuth.de/de/norm/din-en-14351-1/2599166.

[7] RAL-Gütegemeinschaft Fenster und Haustüren e. V. *Leitfaden zur Montage von Fenstern und Haustüren. Edition März 2024*. Edition März 2024, ca. 410 S. 2024.

[8] Jan-Iwo Jäkel et al. "Ontology for a Knowledge-Based Deconstruction of Buildings Based on BIM Models and Linked Data Principles". In: *Buildings* 15.5 (2025), p. 720. DOI: 10.3390/buildings15050720. URL: https://doi.org/10.3390/buildings15050720.

[9] Jerson Alexis Pinzon Amorocho and Timo Hartmann. "Reno-Inst: An ontology to support renovation projects planning and renovation products installation". In: *Advanced Engineering Informatics* 50 (2021), p. 101415. DOI: 10.1016/j.aei.2021.101415. URL: https://doi.org/10.1016/j.aei.2021.101415.

[10] Authors not fully listed in preprint (Cambridge CARES team). "The Material Passport Ontology: A semantic framework for product circularity assessment". In: *Preprint: C4E-328* (2024). Preprint supported by Horizon Europe grant 101058732 and UKRI grant 10038588. URL: https://doi.org/10.17863/CAM.111136.

[11] W3C Linked Building Data Community Group. *The Building Topology Ontology (BOT)*. https://w3c-lbd-cg.github.io/bot/. Public Draft – soliciting public comment. 2021.

[12] Natalya F. Noy and Deborah L. McGuinness. *Ontology Development 101: A Guide to Creating Your First Ontology*. Tech. rep. KSL-01-05. Available at https://protege.stanford.edu/publications/ontology_development/ontology101.pdf. Stanford Knowledge Systems Laboratory and Stanford Medical Informatics, 2001.